Real-time Video Segmentation System Based on FPGA with Frame Buffer

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***Abstract*—With the rapid advancement of artificial intelligence, research in computer vision has deepened significantly. Video image segmentation, a core task in this field, has seen numerous algorithms proposed in recent years. Most existing methods rely on costly hardware platforms like CPUs and GPUs, making large-scale deployment challenging. To address this, we explore FPGA as a high-speed, low-power alternative for implementing video image segmentation. This paper focuses on the FPGA-based simulation of two key algorithms: an improved Sobel edge detection method and the Gaussian Mixture Model (GMM). Based on the frame buffer, we propose an improved Sobel algorithm that adopts a mechanism similar to "majority voting". Both algorithms are implemented using Verilog on the Vivado platform, demonstrating their feasibility for video image segmentation. The results validate the potential of FPGA for efficient and scalable video segmentation systems, offering a foundation for future real-time applications.**

Keywords—Computer Vision, Image Segmentation, FPGA，Sobel，GMM

I. Introduction

The field of computer vision currently encompasses four core tasks: image classification, object detection, object recognition, and image segmentation [1]. With groundbreaking advancements in parallel computing technologies, machine learning algorithms, neural network architectures, and other hardware and software technologies, the image processing capabilities of computers have significantly improved, and the application scope of computer vision technology continues to expand [2]. Among these, video image segmentation, as an important research direction in the field of computer vision, can essentially be seen as an extension and deepening of object detection tasks [3]. Through advanced object detection algorithms, it is possible to effectively identify key information such as moving objects and anomalous entities, thereby achieving precise image segmentation results.

In certain key fields such as security field, if the anomalies can be detected quickly, the performance of the security system will be significantly enhanced, effectively reducing potential security risks. However, current video image segmentation systems are primarily deployed on traditional computing platforms such as CPUs and GPUs, which suffer from inherent limitations such as insufficient real-time performance and high energy consumption. Moreover, deploying video segmentation systems on mobile devices like drones presents additional challenges due to constrained computational resources and limited storage space [4].

To address these issues, implementing video segmentation algorithms on high-speed, low-power FPGA (Field-Programmable Gate Array) platforms has emerged as a highly promising technical solution [5]. FPGAs, with their strong parallel computing capabilities, high energy efficiency, and excellent reconfigurability, can effectively meet the real-time and low-power requirements of video segmentation algorithms [6]. By using FPGA to complete the image processing algorithm, the processing efficiency can be greatly improved [7]. Integrating FPGAs with video segmentation algorithms not only enhances the overall performance of video segmentation systems but also promotes the widespread application of this technology in everyday scenarios, offering significant practical value and social benefits.

The main work of this paper focuses on the FPGA-based implementation of an improved Sobel algorithm and a Gaussian Mixture Model (GMM) algorithm for offline video segmentation. Compared to the traditional Sobel algorithm, the improved Sobel algorithm introduces frame buffering and a voting mechanism, which significantly enhances the smoothness of video frames and reduces noise during video stream processing. Specifically, this paper first simulates a single-frame video image segmentation method on an FPGA using the improved Sobel algorithm and compares the simulation results with those of the traditional Sobel algorithm.

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Subsequently, the FPGA-based simulation of the GMM algorithm for video image segmentation is also implemented.

Section 2 and Section 3 describe the improved Sobel algorithm and GMM algorithm respectively, and provide details of the FPGA simulation experiments. Section 4 conclude the paper.

II. Improved Sobel Algorithm Based on Frame Buffer

A. Algorithm Overview

The Sobel algorithm is primarily used for detecting edges in images, where edges typically correspond to points with significant changes in grayscale intensity. The Sobel operator combines Gaussian smoothing with differentiation operations to compute gradients in both the horizontal and vertical directions of an image. The greater the change in grayscale intensity, the larger the first derivative value. Compared to the Canny operator, the Sobel operator, while less accurate in edge detection [8], offers higher computational efficiency and stronger noise resistance.

However, when applied to video streams, the traditional Sobel algorithm can produce noisy results due to rapid changes between frames. To address this, we introduce a "majority voting" mechanism that leverages the temporal continuity of video frames.

B. Algorithm Improvement

The improved Sobel algorithm incorporates a frame buffer to store the segmentation results of the previous two frames. For each pixel, a voting mechanism is applied: if two or more of the current, previous, and second-previous frames classify the pixel as an edge, it is marked as an edge in the current frame. This approach reduces noise and produces smoother segmentation results.

The algorithm consists of three main steps [9]:

1.Horizontal and Vertical Gradient Calculation: The Sobel operator calculates the gradient in both horizontal and vertical directions using convolution kernels.

$$G_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix} * A \tag{1}$$

$$G_y = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{bmatrix} * A \tag{2}$$

In the formula, A represents a 3x3 matrix composed of the current pixel and its surrounding neighboring elements.

2.Gradient Magnitude Calculation: The gradient magnitude is computed using the Euclidean norm of the horizontal and vertical gradients.

$$G = \sqrt{{G_x}^2 + {G_y}^2} \tag{3}$$

3.Thresholding and Voting: A threshold is applied to the gradient magnitude to classify pixels as edges or non-edges. The voting mechanism is then applied to reduce noise.

The overall procedure is shown in Fig.1 below:

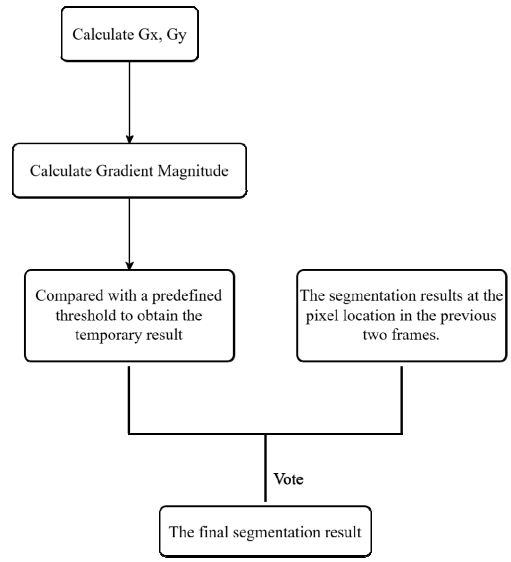

Fig. 1. Overall improved Sobel algorithm procedure.

C. FPGA Simulation

In this paper, the image processing module designed for the simulation of the Sobel algorithm-based image segmentation consists of three components: image grayscale conversion, median filtering, and the Sobel algorithm.

When the improved Sobel algorithm is used for image segmentation simulation, two cases of edge segmentation threshold of 64 and 128 are selected for simulation, and the segmentation result of the improved algorithm is compared with that of the original algorithm.

The image selected by the simulation is the 10th frame in the video stream, Fig.2 is the original video image, Fig.3 is the segmentation image when the threshold is 64, and Fig.4 is the segmentation image when the threshold is 128. In Fig.3 and Fig.4, the left figure is the image segmentation result of the original Sobel algorithm, and the right is the image segmentation result after using the improved Sobel algorithm.

Fig. 2. Original video image.

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Fig. 3. Segmentation image when the threshold is 64

Fig. 4. Segmentation image when the threshold is 128

Compared with the left and right images in Fig.3 and Fig.4, it can be observed that after applying the improved Sobel algorithm, the segmented image exhibits fewer noise artifacts compared to the results from the original Sobel algorithm. The overall image quality is smoother, and the smoothing effect of the improved algorithm becomes more pronounced when the edge segmentation threshold of the Sobel algorithm is set to a smaller value.

In terms of algorithm processing speed, when using FPGA to process 30 frames of video image with a resolution of 640*480, it can be seen from the simulation that it only takes 184.6ms to complete the processing, which can well meet the real-time requirements of the video segmentation system.

III. Gaussian Mixture Model (GMM) Algorithm

A. Algorithm Overview

The GMM (Gaussian Mixture Model) algorithm, or Gaussian Mixture Model algorithm, is a statistical method used for modeling complex backgrounds. Under mathematical experimental conditions, it has been proven that various factors in natural environments (such as wind speed and lighting) follow a Gaussian distribution [10]. Therefore, a single Gaussian distribution can effectively model a static background. However, in real-world scenarios, backgrounds are often complex and dynamic, making it difficult for a single Gaussian distribution to accurately represent such variability. To address this, researchers proposed using a weighted mixture of multiple Gaussian distributions to model complex backgrounds. In this approach, each pixel is represented by a mixture of Gaussian distributions, and the model is continuously updated over time to adapt to changes in the scene. The Gaussian Mixture Model provides an excellent fit for complex environments, offering strong robustness, and is particularly well-suited for background modeling tasks.

The implementation of the GMM (Gaussian Mixture Model) algorithm requires the following steps[11]:

1. Background Parameter Initialization: Initialize the parameters of the Gaussian distributions for each pixel in the background model.

2. Model Matching and Updating Gaussian Distribution Parameters: Match the current pixel values with the existing Gaussian distributions and update the parameters (mean, variance, and weight) of the distributions based on the match results.

3. Foreground Segmentation for Input Video Frames: Use the updated Gaussian mixture model to classify pixels as either background or foreground, thereby achieving foreground segmentation.

Due to the need to maintain and update multiple Gaussian distributions for each pixel, the GMM algorithm involves significant computational complexity and consumes substantial resources.

The overall procedure is shown in Fig.5 below:

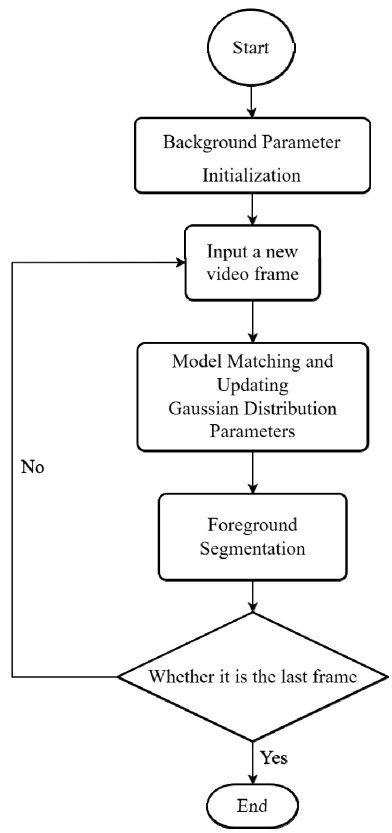

Fig. 5. Overall GMM algorithm procedure.

B. Background model initialization and update strategy

The Gaussian mixture model is composed of multiple single Gaussian models. The mathematical expression of a single Gaussian model is as follows:

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) \tag{4}$$

And the mathematical expression of Gaussian mixture model is:

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$$G(x_t) = \sum_{k=1}^{K} \omega_{k,t} N_k(X_t, \mu_{k,t}, \sigma_{k,t}) \quad (5)$$

In formula (5), K is the total number of Gaussian distributions, $\omega_{k,t}$ is the weight of the K single Gaussian distribution at time t, and N_k is the K single Gaussian distribution function.

As can be seen from the above mathematical expressions, the initialization of the Gaussian mixture model is mainly to assign values to ω_k,μ_k,σ_k, that is, the weight, mean value and standard deviation of each single Gaussian distribution. The common initialization methods include choosing the first few frames of the video stream to do the average calculation of the assignment method, and directly assigning the initial values of each Gaussian distribution parameter. In this paper, the initial value of each Gaussian distribution parameter is directly assigned. In this simulation experiment, K is set to 4, with initial mean values of 0, 85, 170, and 255, respectively. The initial variance is uniformly set to 100, and the weights are uniformly set to 0.25.

Whether the newly input pixel points match the background model is mainly determined by the following two mathematical formulas [12]:

$$\left|X_t - \mu_{k,t-1}\right| < 6.25 * \sigma_{k,t-1} \quad (6)$$

$$\left|X_t - \mu_{k,t-1}\right| < D \quad (7)$$

In formula (6), X_t represents the newly input pixel, $\mu_{k,t-1}$ represents the mean of the current background model, and $\sigma_{k,t-1}$ represents the standard deviation of the current background model. In formula (7), D represents the pre-set matching threshold. If either of the above two formulas is satisfied, the pixel can be identified as background; otherwise, it is foreground.

When a pixel is identified as background, the update of the Gaussian distribution parameters will follow the following rules:

$$\begin{cases} \omega_{k,t} = (1 - \alpha)\omega_{k,t-1} + \alpha \\ \beta = \frac{\alpha}{\omega_{k,t}} \\ \mu_{k,t} = (1 - \alpha)\mu_{k,t-1} + \alpha X_t \\ {\sigma_{k,t}}^2 = (1 - \beta){\sigma_{k,t-1}}^2 + \beta(X_t - \mu_{k,t-1})^2 \end{cases} \quad (8)$$

In formula (8), α is the learning rate, and β represents the change rate of each parameter of the model. In this paper, the learning rate is set to 0.125. If the pixel is identified as foreground, then only the weights need to be updated:

$$\omega_{k,t} = (1 - \alpha)\omega_{k,t-1} \quad (9)$$

C. FPGA Simulation

Since decimal operations are not convenient in FPGA, in order to implement the GMM algorithm on FPGA, we use fixed-point number to represent parameters such as weights, so that they can be efficiently calculated in FPGA.

The video image processing module designed in this paper consists of image grayscale conversion, image median filtering, the GMM algorithm, and image morphological processing. The simulation results based on FPGA are as follows：

As shown in Fig.6, the left picture is the 50th frame of the selected video, and the right picture is the 50th frame of the image after segmentation by the image processing module. It can be seen that the algorithm can recognize moving objects and achieve the purpose of segmentation of foreground and background.

Fig. 6. Image segmentation results of GMM algorithm.

When compared with the ground truth created by Opencv, it can be found that the simulation result of the FPGA still shows some discrepancies compared to the processing results on the PC side.The figure and the table are shown below:

Fig. 7. Comparison between simulation result and ground truth.

TABLE I. PERFORMANCE METRICS

Metrics	***Precision***	***Recall***	***F-Score***
Value	0.5645	0.4646	0.5097

During the simulation of the entire image processing module, it was observed that the FPGA can complete image segmentation for 60 frames of video with a resolution of 640*480 in just 184ms. This performance effectively meets the system's requirements for real-time processing.

IV. CONCLUSION

This paper presents the principles of improved Sobel and GMM algorithms and performs FPGA-based simulations for both. Both algorithms can effectively segment video images, and the improved Sobel algorithm based on frame buffer has

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more advantages in video smoothness compared with the traditional Sobel algorithm.

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